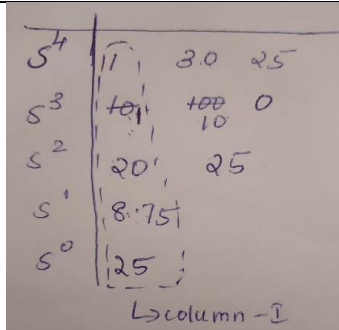
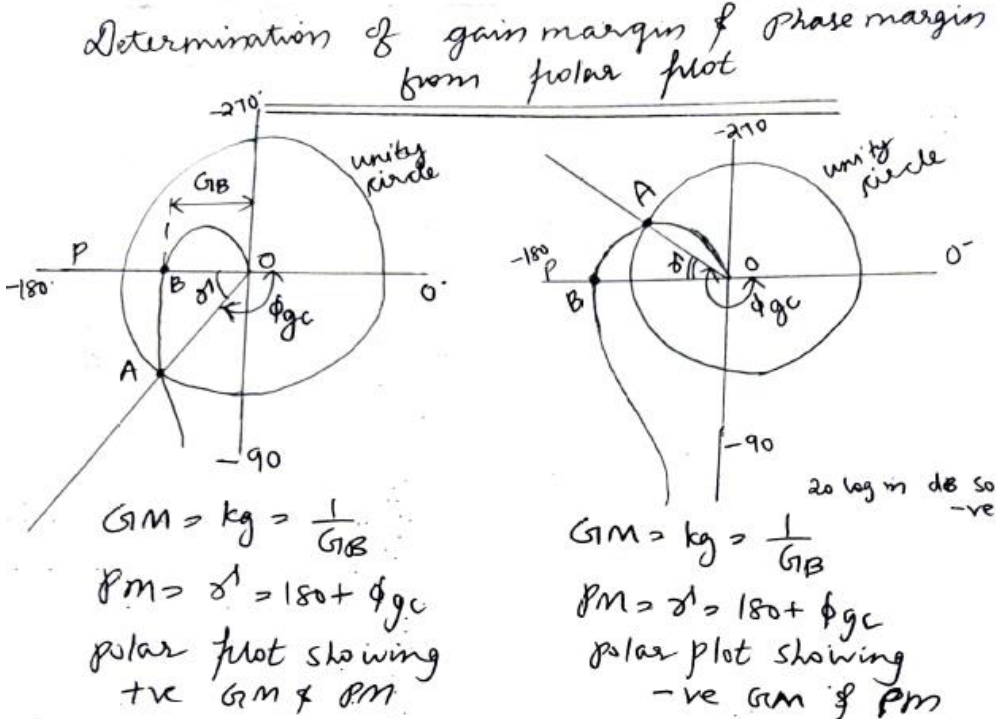
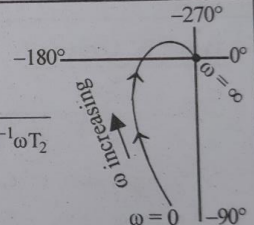


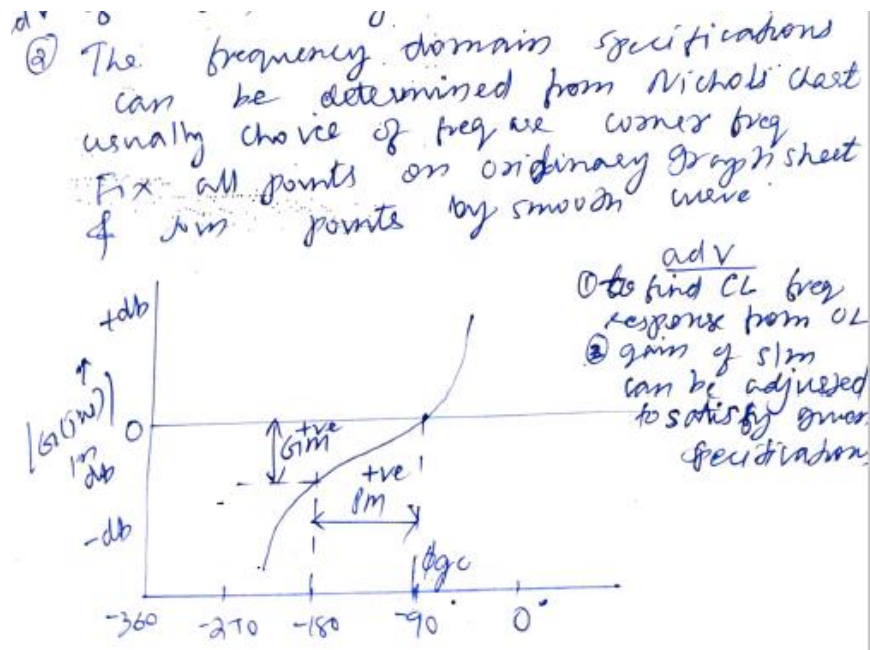
Sahrdaya College of Engineering and Technology, Kodakara		
Answer Key		
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY SIXTH SEMESTER B.TECH DEGREE EXAMINATION, JUNE 2022		
Course Code: EET302		
Course Name: LINEAR CONTROL SYSTEMS		
PART A		
		<i>Answer all questions. Each question carries 3 marks</i>
	Q1	Explain the effect of feedback on overall gain, sensitivity and noise for a closed loop control system.
	Ans	For negative feedback - Overall gain will be reduced - Sensitivity will be reduced - Noise rejection can be achieved
	Q2	Explain the role of compensators in automatic control systems.
	Ans	Role of compensators in automatic control systems- 1. To stabilize the systems 2. To achieve the desired performance in case of a stable system 3. Lag compensator for reducing the steady state error and lead compensators for faster response
	Q3	What is meant by bounded input bounded output stability?
	Ans	Bounded input bounded output stability – Definition – for a bounded input output also bounded Condition – Impulse response should be absolutely summable
	Q4	Using Routh's stability criterion determine the stability of the given system whose characteristic equation is $s^4 + 10s^3 + 30s^2 + 100s + 25 = 0$
	Ans	

		On examining the elements of first column of routh array it is observed that all elements are positive and there is no sign change. Hence all roots are lying on the left half of s-plane and the system is stable.
Q5		Compare the performance characteristics of PI and PD controllers.
Ans	PI Controllers	It is a device that produces an o/p signal consisting of 2 terms one proportional to i/p signal $e(t)$ & other proportional to integral of i/p signal $e(t)$ where k_p is proportional gain & k_i is integral gain constant $u(t) = e(t) + \int e(t) dt$ $= k_p e(t) + k_i \int e(t) dt$ $u(s) = k_p E(s) + k_i \frac{E(s)}{s}$ $TF = k_p + \frac{k_i}{s}$ A PI controller helps in reducing both the rise time and the steady state errors of the system. PD Controllers Proportional plus derivative $u(t) \propto e(t) + \frac{d}{dt} e(t)$ $= k_p e(t) + k_d \frac{d}{dt} e(t)$ $u(s) = k_p E(s) + k_d s E(s)$ $TF = k_p + s k_d$ A PD controller reduces the transients like rise time, overshoot, and oscillations in the output.
Q6		Explain the effect of adding poles and zeroes to the nature of root locus

Ans	Effect of adding poles 1. Root locus shift towards imaginary axis 2. Relative stability decreases 3. Settling time increases Effect of adding zeroes 1. Root locus shift towards left of s plane 2. Relative stability increases 3. Settling time decreases
Q7	Explain how phase margin and gain margin can be detected from the polar plot
Ans	Determination of gain margin & phase margin from polar plot  $GM = k_g = \frac{1}{G_B}$ $PM = \delta' = 180 + \phi_{gc}$ polar plot showing +ve GM & PM $GM = k_g = \frac{1}{G_B}$ $PM = \delta' = 180 + \phi_{gc}$ polar plot showing -ve GM & PM 20 log m dB so -ve
Q8	Draw the appropriate polar plot for the open loop transfer function, $G(s) = \frac{1}{s(1+sT_1)(1+sT_2)}$
Ans	

		Type : 1, Order : 3 $G(s) = \frac{1}{s(1+sT_1)(1+sT_2)}$ $G(j\omega) = \frac{1}{j\omega(1+j\omega T_1)(1+j\omega T_2)} = \frac{1}{\omega \angle 90^\circ \sqrt{1+\omega^2 T_1^2} \angle \tan^{-1} \omega T_1 \sqrt{1+\omega^2 T_2^2} \angle \tan^{-1} \omega T_2}$ $= \frac{1}{\omega \sqrt{(1+\omega^2 T_1^2)(1+\omega^2 T_2^2)}} \angle (-90^\circ - \tan^{-1} \omega T_1 - \tan^{-1} \omega T_2)$ As $\omega \rightarrow 0$, $G(j\omega) \rightarrow \infty \angle -90^\circ$ As $\omega \rightarrow \infty$, $G(j\omega) \rightarrow 0 \angle -270^\circ$ 
Q9	State and explain Nyquist stability criterion.	
Ans	Nyquist stability criterion	"If the $G(s)H(s)$ contour in the $G(s)H(s)$-plane corresponding to Nyquist contour in the s-plane encircles the point $-1 + j0$ in the anticlockwise direction as many times as the number of right half s-plane poles of $G(s)H(s)$, then the closed loop system is stable". In examining the stability of linear control systems using the Nyquist stability criterion, we come across the following three situations. No encirclement of $-1 + j0$ point : This implies that the system is stable if there are no poles of $G(s)H(s)$ in the right half s-plane. If there are poles on right half s-plane then the system is unstable. Anticlockwise encirclements of $-1 + j0$ point : In this case the system is stable if the number of anticlockwise encirclements is same as the number of poles of $G(s)H(s)$ in the right half s-plane. If the number of anticlockwise encirclements is not equal to number of poles on right half s-plane then the system is unstable. Clockwise encirclements of the $-1 + j0$ point : In this case the system is always unstable. Also in this case, if no poles of $G(s)H(s)$ in right half s-plane, then the number of clockwise encirclement is equal to number of poles of closed loop system on right half s-plane.
Q10	Explain the significance of Nichols chart	
Ans	Nichols chart	Nichols chart consists of M & N contours. The M contours are the magnitude of the CL s/m in decibels & N contours are the corresponding phase angle locus of CL s/m. Nichols plot of CL s/m can be plotted on the ordinary graph. The Nichols plot is a graph b/w magnitude of $G(j\omega)$ in db along Y axis and phase $\angle$ of $G(j\omega)$ along X axis for various ω values.

Advantages



PART B

Answer any one full question from each module. Each question carries 14 marks

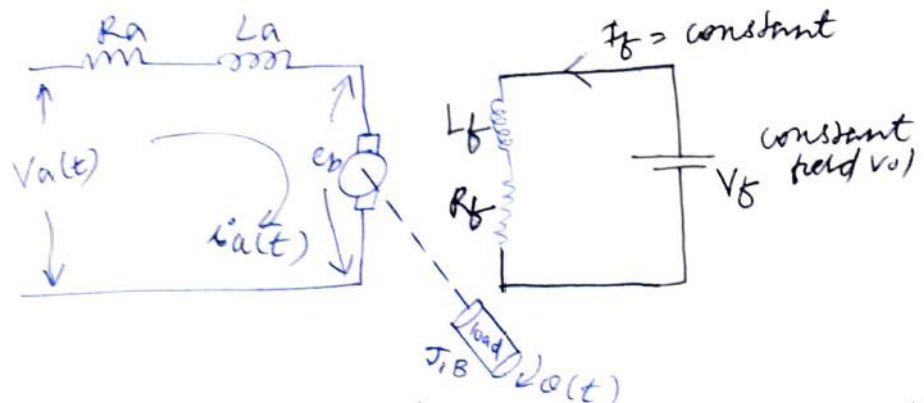
Module 1

Q11

- Derive the transfer function for the armature controlled DC motor and hence draw the block diagram representation of the system
- Obtain the transfer function of lag compensators using R-C circuit components and bring out the characteristics of lag compensators

Ans

- The transfer function for the armature controlled DC motor



Expression for torque, armature voltage and mechanical equation

k_t - Torque constant

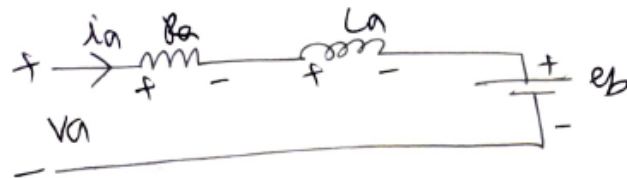
T - torque developed by motor

θ - angular displacement of shaft

J - moment of inertia

B - frictional coefficient

k_b - back emf constant



KVL $I_a R_a + L_a \frac{dI_a}{dt} + E_b = V_a$

$T \propto \phi I_a$ (DC motor)

Since flux constant here

$T \propto I_a$ $T = k_t I_a$

Differential eqn governing mechanical s/m

$$J \frac{d^2 \theta}{dt^2} + B \frac{d\theta}{dt} = T$$

Back emf of dc machine proportional to speed (angular velocity) of shaft.

$$E_b \propto \frac{d\theta}{dt} \quad E_b = k_b \frac{d\theta}{dt}$$

Taking LT of eqns

$$I_a(s) R_a + L_a s I_a(s) + E_b(s) = V_a(s) \quad \text{--- (1)}$$

$$T(s) = k_t I_a(s) \text{ --- (2)}$$

$$Js^2 \theta(s) + Bs \theta(s) = T(s) \text{ --- (3)}$$

$$E_b(s) = k_b s \theta(s) \text{ --- (4)}$$

from (2) & (3)

$$k_t I_a(s) = (Js^2 + Bs) \theta(s)$$

$$\text{i.e. } \theta(s) = \frac{k_t I_a(s)}{Js^2 + Bs} \text{ --- (i)}$$

put (i) in (3)

$$Js^2 \theta(s) + Bs \theta(s) = k_t I_a(s)$$

$$I_a(s) = \frac{(Js^2 + Bs) \theta(s)}{k_t} \text{ --- (5)}$$

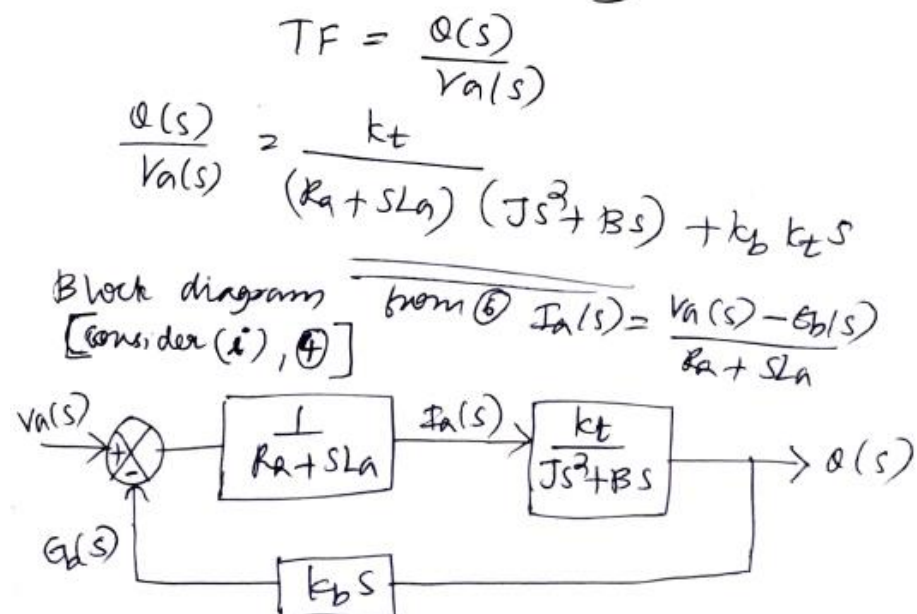
$$\text{Eqn (1) gives } (R_a + sL_a) I_a(s) + E_b(s) = V_a(s)$$

$$\text{Put (4) \& (5) in (6) --- (6)}$$

$$(R_a + sL_a) \left(\frac{Js^2 + Bs}{k_t} \theta(s) \right) + k_b s \theta(s) = V_a(s)$$

$$\left[\frac{(R_a + sL_a)(Js^2 + Bs)}{k_t} + k_b s \right] \theta(s) = V_a(s)$$

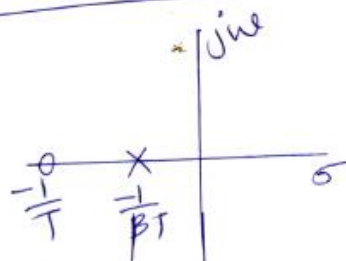
$$\left[\frac{(R_a + sL_a)(Js^2 + Bs) + k_b k_t s}{k_t} \right] \theta(s) = V_a(s)$$



The block diagram representation of the system

b) Lag compensator

s plane representation

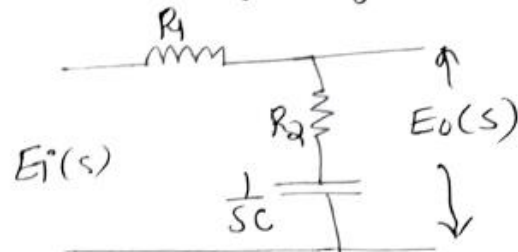


lag compensator has pole at $s = -\frac{1}{\beta T}$
and zero at $s = -\frac{1}{T}$

$$TF G_c(s) = \frac{s + z_c}{s + p_c} = \frac{s + \frac{1}{T}}{s + \frac{1}{\beta T}} \quad \text{--- (1)}$$

$$T > 0 ; \beta > 1$$

Realisation of lag compensator using RC n/w



$$E_i(s) = \text{input vol} \quad E_o(s) = \text{output vol}$$

$$E_o(s) = E_i(s) \frac{R_2 + \frac{1}{sC}}{R_1 + R_2 + \frac{1}{sC}} = E_i(s) \frac{(1 + sR_2C)}{sC(R_1 + R_2) + 1}$$

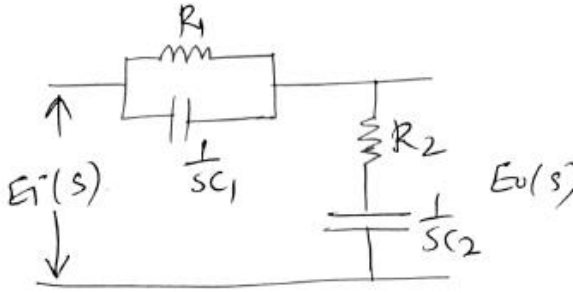
$$TF = \frac{E_o(s)}{E_i(s)} = \frac{C R_2 (s + \frac{1}{C R_2})}{C(R_1 + R_2) [s + \frac{1}{C(R_1 + R_2)}]}$$

$$= \frac{s + \frac{1}{R_2 C}}{s + \frac{1}{(\frac{R_1 + R_2}{R_2}) R_2 C}} \quad \text{--- (2)}$$

comparing ① & ②

$$\frac{E_o(s)}{E_i(s)} = \frac{1}{\beta} \frac{(s + \frac{1}{T})}{(s + \frac{1}{\beta T})} \quad \text{where } T = R_2 C, \quad \beta = \frac{R_1 + R_2}{R_2}$$

TF of RC n/w is now similar to the TF of a lag compensator with an attenuation of $\frac{1}{\beta}$ since $\beta > 1, \frac{1}{\beta} < 1$ if attenuation not required an amp with gain β can be connected in cascade with RC n/w to nullify the attenuation

Q12	a) Derive the transfer function of a lag-lead compensator b) Derive the transfer function for the field controlled DC motor and hence draw the block diagram representation of the system
Ans	a) Circuit diagram of lag-lead compensator  $\frac{E_o(s)}{E_i(s)} = \frac{R_2 + \frac{1}{sC_2}}{R_1 \cdot \frac{1}{sC_1} + R_2 + \frac{1}{sC_2}}$ $= \frac{sR_2C_2 + 1}{\frac{R_1}{sR_1C_1 + 1} + \frac{sR_2C_2 + 1}{sC_2}}$ $= \frac{(sR_2C_2 + 1)(sR_1C_1 + 1)}{sR_1C_2 + (sR_2C_2 + 1)(sR_1C_1 + 1)}$ $= \frac{R_1R_2C_1C_2 \left(s + \frac{1}{R_2C_2}\right) \left(s + \frac{1}{R_1C_1}\right)}{sR_1C_2 + R_1R_2C_1C_2 \left(s + \frac{1}{R_2C_2}\right) \left(s + \frac{1}{R_1C_1}\right)}$

$$\begin{aligned}
 & \frac{\left(s + \frac{1}{R_2 C_2}\right) \left(s + \frac{1}{R_1 C_1}\right)}{= \frac{s}{R_2 C_1} + \left(s + \frac{1}{R_2 C_2}\right) \left(s + \frac{1}{R_1 C_1}\right)} \\
 & = \frac{\left(s + \frac{1}{R_2 C_2}\right) \left(s + \frac{1}{R_1 C_1}\right)}{s^2 + s \left(\frac{1}{R_2 C_1} + \frac{1}{R_2 C_2} + \frac{1}{R_1 C_1}\right) + \frac{1}{R_1 R_2 C_1 C_2}} \\
 G_c(s) &= \frac{\left(s + \frac{1}{T_1}\right) \left(s + \frac{1}{T_2}\right)}{s^2 + s \left(\frac{1}{\beta T_1} + \frac{1}{\alpha T_2}\right) + \frac{1}{\alpha \beta T_1 T_2}}
 \end{aligned}$$

comparing

$$\begin{aligned}
 T_1 &= R_1 C_1 \\
 T_2 &= R_2 C_2
 \end{aligned}$$

$$R_1 R_2 C_1 C_2 = \alpha \beta T_1 T_2$$

$$\frac{1}{R_1 C_1} + \frac{1}{R_2 C_2} + \frac{1}{R_2 C_1} = \frac{1}{\beta T_1} + \frac{1}{\alpha T_2}$$

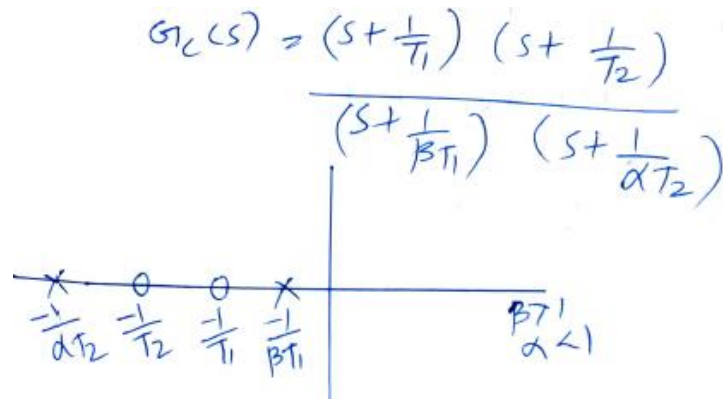
$$\alpha \beta = \frac{R_1 R_2 C_1 C_2}{T_1 T_2}$$

$$\underline{\underline{\alpha \beta = 1}}$$

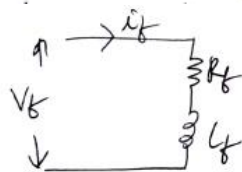
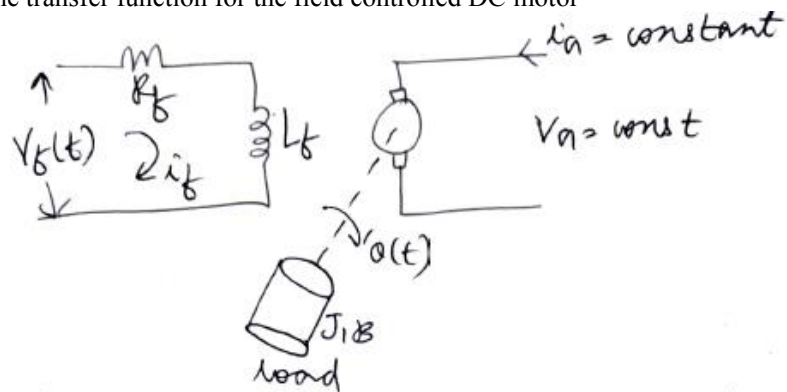
$$G_c(s) = \frac{\alpha \beta (1 + s T_1) (1 + s T_2)}{(1 + s \beta T_1) (1 + s \alpha T_2)}$$

$$\begin{aligned}
 G_c(j\omega) &= \frac{(1 + j\omega T_1) (1 + j\omega T_2)}{(1 + j\omega \beta T_1) (1 + j\omega \alpha T_2)} \\
 \text{corner freq} \quad \omega_{c1} &= \frac{1}{\beta T_1} \quad \omega_{c2} = \frac{1}{T_1} \quad \omega_{c3} = \frac{1}{T_2} \quad \omega_{c4} = \frac{1}{\alpha T_2}
 \end{aligned}$$

Pole-Zero representation



b) The transfer function for the field controlled DC motor



$$\text{KVL } R_f i_f + L_f \frac{di_f}{dt} = V_f$$

$T \propto \phi i_a$ here i_a constant

$T \propto \phi$ but $\phi \propto i_f$ so $T \propto i_f$

$T = k_f i_f$ where k_f some const

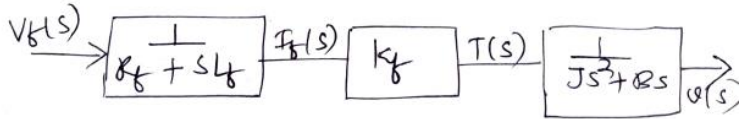
Differential eqn of mechanical s/m

$$T = J \frac{d^2\theta}{dt^2} + B \frac{d\theta}{dt}$$

Taking LT $R_f I_f(s) + L_f s I_f(s) = V_f(s) \rightarrow ①$

$$T(s) = k_f I_f(s) \rightarrow ②$$

$$J s^2 \theta(s) + B s \theta(s) = T(s) \rightarrow ③$$

		from ② & ③ $Js^2 \theta(s) + Bs \theta(s) = k_f I_f(s)$ $I_f(s) = \frac{V_f(s)}{R_f + sL_f} \quad \text{--- ④}$ put in ③ $k_f \frac{V_f(s)}{R_f + sL_f} = (Js^2 + Bs) \theta(s)$ $\frac{\theta(s)}{V_f(s)} = \frac{k_f}{(Js^2 + Bs)(R_f + sL_f)}$ Block diagram ① $\Rightarrow I_f(s) = \frac{V_f(s)}{R_f + sL_f}$ ② $\Rightarrow T(s) = k_f I_f(s)$ from ② & ③ $\theta(s) = \frac{T(s)}{Js^2 + Bs}$ 
Module 2		
Q13	a)	The forward path transfer function of a unity feedback control system is given by $G(s) = \frac{2}{s(s+3)}$ Obtain the unit step response of the system b) Derive the error coefficient and steady state error for a type 1 system applied with unit step and unit ramp input
Ans	a)	The forward path transfer function of a unity feedback control system is given by $G(s) = \frac{2}{s(s+3)}$

C.L.T.F

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}$$

$$= \frac{2}{s(s+3)} \cdot \frac{1}{1 + \frac{2}{s(s+3)}}$$

$$= \frac{2}{s(s+3)+2} = \frac{2}{s^2+3s+2}$$

$$= \frac{2}{(s+1)(s+2)}$$

i/p is unit step

$$R(s) = \frac{1}{s}$$

$$C(s) = \frac{2}{s(s+1)(s+2)}$$

$$C(s) = \frac{2}{s(s+1)(s+2)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2}$$

$$A = C(s) \times s \big|_{s=0} = 1$$

$$B = C(s) (s+1) \big|_{s=-1} = -2$$

$$C = 1$$

$$C(s) = \frac{1}{s} - \frac{2}{s+1} + \frac{1}{s+2}$$

$$C(t) = 1 - 2e^{-t} + e^{-2t}$$

- b) Derive the error coefficient and steady state error for a type 1 system with general transfer function of type 1 system

Unit step signal

$$\text{Steady state error, } e_{ss} = \lim_{s \rightarrow 0} \frac{sR(s)}{1+G(s)H(s)}$$

When the input is unit step, $R(s) = 1/s$

$$\therefore e_{ss} = \lim_{s \rightarrow 0} \frac{s \cdot \frac{1}{s}}{1+G(s)H(s)} = \lim_{s \rightarrow 0} \frac{1}{1+G(s)H(s)} = \frac{1}{1 + \lim_{s \rightarrow 0} G(s)H(s)} = \frac{1}{1+K_p}$$

$$\text{where, } K_p = \lim_{s \rightarrow 0} G(s)H(s)$$

The constant K_p is called positional error constant.

		<u>Type-1 system</u> $K_p = \lim_{s \rightarrow 0} G(s)H(s) = \lim_{s \rightarrow 0} K \frac{(s+z_1)(s+z_2)(s+z_3)\dots}{s(s+p_1)(s+p_2)(s+p_3)\dots} = \infty$ $\therefore e_{ss} = \frac{1}{1+K_p} = \frac{1}{1+\infty} = 0$ In systems with type number 1 and above, for unit step input the value of K_p is in steady state error is zero. Unit Ramp Signal Steady state error, $e_{ss} = \lim_{s \rightarrow 0} \frac{sR(s)}{1+G(s)H(s)}$ When the input is unit ramp, $R(s) = \frac{1}{s^2}$ $\therefore e_{ss} = \lim_{s \rightarrow 0} \frac{s \cdot \frac{1}{s^2}}{1+G(s)H(s)} = \lim_{s \rightarrow 0} \frac{1}{s+G(s)H(s)} = \frac{1}{\lim_{s \rightarrow 0} sG(s)H(s)} = \frac{1}{K_v}$ where, $K_v = \lim_{s \rightarrow 0} sG(s)H(s)$ The constant K_v is called velocity error constant. <u>Type-1 system</u> $K_v = \lim_{s \rightarrow 0} sG(s)H(s) = \lim_{s \rightarrow 0} sK \frac{(s+z_1)(s+z_2)(s+z_3)\dots}{s(s+p_1)(s+p_2)(s+p_3)\dots} = K \frac{z_1 \cdot z_2 \cdot z_3 \dots}{p_1 \cdot p_2 \cdot p_3 \dots} = \text{constant}$ $\therefore e_{ss} = 1/K_v = \text{constant}$ Hence in type-1 systems when the input is unit ramp there will be a constant steady state error.
Q14	a) The open loop transfer function of a unity feedback control system is given by $G(s) = \frac{25}{s(s+5)}$ Determine the natural frequency of oscillation, damped frequency of oscillation, damping ratio and maximum overshoot for a unit step input. b) Obtain the unit step response of a standard first order system. Draw the response curve and comment on the effect of time constant on the speed of response of the system	
Ans	a) The open loop transfer function of a unity feedback control system is given by $G(s) = \frac{25}{s(s+5)}$	

$$\begin{aligned}
 CLTF &= \frac{4(8)}{1+4(8)} \\
 &= \frac{25}{s(s+5)} = \frac{25}{s(s+5)+25} \\
 &= \frac{25}{s^2+5s+25}
 \end{aligned}$$

Second order system general equation

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} = \frac{25}{s^2 + 5s + 25}$$

$$\omega_n = 5 \text{ rad/sec}$$

Damped frequency of oscillations

$$2\xi\omega_n = 5$$

$$\xi = \frac{5}{10} = 0.5$$

$$\omega_d = \omega_n \sqrt{1-\xi^2} = 5\sqrt{1-(0.5)^2} = 4.33 \text{ rad/sec}$$

$$\therefore M_p = e^{-\xi\pi/\sqrt{1-\xi^2}} \times 100 = 16.3\%$$

b) Transfer function of a standard first order system

3.5 RESPONSE OF FIRST ORDER SYSTEM FOR UNIT STEP INPUT

The closed loop order system with unity feedback is shown in fig 3.6.

$\Rightarrow R(s) \rightarrow \left[\frac{1}{1+Ts} \right] \rightarrow C(s)$

Fig 3.6 : Closed loop first order system.

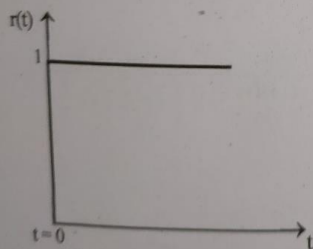
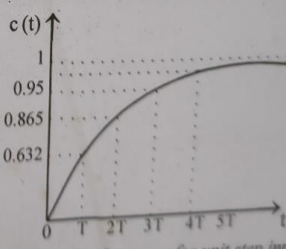
The closed loop transfer function of first order system, $\frac{C(s)}{R(s)} = \frac{1}{1+Ts}$

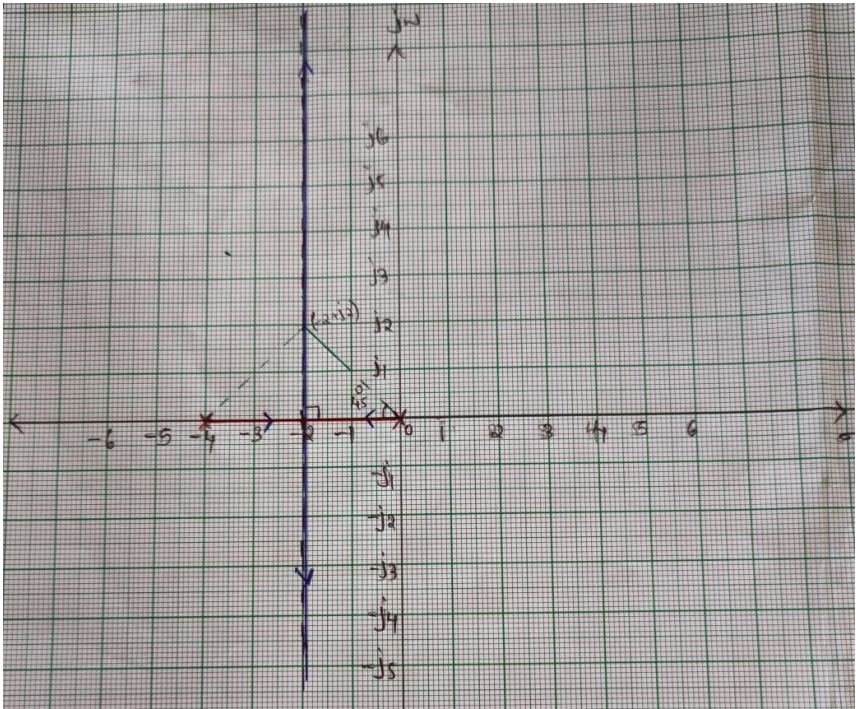
If the input is unit step then, $r(t) = 1$ and $R(s) = \frac{1}{s}$.

$\therefore$ The response in s-domain, $C(s) = R(s) \frac{1}{(1+Ts)} = \frac{1}{s} \frac{1}{(1+Ts)} = \frac{1}{sT \left(\frac{1}{T} + s \right)} = \frac{\frac{1}{T}}{s \left(s + \frac{1}{T} \right)}$

By partial fraction expansion,

$$C(s) = \frac{\frac{1}{T}}{s \left(s + \frac{1}{T} \right)} = \frac{A}{s} + \frac{B}{s + \frac{1}{T}}$$

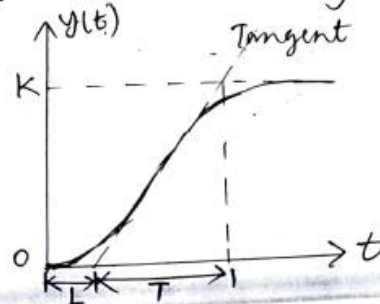
		A is obtained by multiplying C(s) by s and letting $s = 0$. $A = C(s) \times s \Big _{s=0} = \frac{\frac{1}{T}}{s + \frac{1}{T}} \times s \Big _{s=0} = \frac{\frac{1}{T}}{\frac{1}{T}} = 1$ B is obtained by multiplying C(s) by $(s+1/T)$ and letting $s = -1/T$. $B = C(s) \times \left(s + \frac{1}{T}\right) \Big _{s=-\frac{1}{T}} = \frac{\frac{1}{T}}{s + \frac{1}{T}} \times \left(s + \frac{1}{T}\right) \Big _{s=-\frac{1}{T}} = \frac{\frac{1}{T}}{-\frac{1}{T}} = -1$ $\therefore C(s) = \frac{1}{s} - \frac{1}{s + \frac{1}{T}}$ The response in time domain is given by, $c(t) = \mathcal{L}^{-1}\{C(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s} - \frac{1}{s + \frac{1}{T}}\right\} = 1 - e^{-\frac{t}{T}} \quad \dots(3.13)$ The equation (3.13) is the response of the closed loop first order system for unit step input. For step input of step value, A, the equation (3.13) is multiplied by A. $\therefore$ For closed loop first order system, Unit step response $= 1 - e^{-\frac{t}{T}}$ Step response $= A \left(1 - e^{-\frac{t}{T}}\right)$ When, $t = 0$, $c(t) = 1 - e^0 = 0$ When, $t = 1T$, $c(t) = 1 - e^{-1} = 0.632$ When, $t = 2T$, $c(t) = 1 - e^{-2} = 0.865$ When, $t = 3T$, $c(t) = 1 - e^{-3} = 0.95$ When, $t = 4T$, $c(t) = 1 - e^{-4} = 0.9817$ When, $t = 5T$, $c(t) = 1 - e^{-5} = 0.993$ When, $t = \infty$, $c(t) = 1 - e^{-\infty} = 1$ Here T is called Time constant of the system. In a time of 5T, the system is assumed to have attained steady state. The input and output signal of the first order system is shown in fig 3.7.  Fig 3.7a : Unit step input.  Fig 3.7b : Response for unit step input.
Module 3		
Q15	a) A unity feedback control system has an open loop transfer function $G(s) = \frac{K}{s(s+4)}$ Draw the root locus and determine the value of K if the damping ratio is 0.707.	

		b) Explain the Ziegler Nichols method of tuning the PID controllers.
Ans	a)	A unity feedback control system has an open loop transfer function $G(s) = \frac{K}{s(s+4)}$ Breakaway point $s=-2$ Angle of asymptotes = 90 and 270 degree Centroid = -2  Line at angle of 45 degree intersect n root locus at $-2+j2$, equate magnitude =1 at this value of s, then $K = 8$
	b)	Ziegler Nichols method of tuning the PID controllers

First method of ZN Tuning for PID controller

Dynamic model of plant not known

Applied to plants with step responses of form shown in fig



Typical step response for overdamped case

TF of plant may be approximated by a first order s/m as

$$\frac{C(s)}{R(s)} = G(s) = \frac{K e^{-LS}}{1 + ST}$$

The values of K_p , T_i & T_d suggested by ZN are given in table

Type of controller	K_p	$T_i = K_p / k_i$	$T_d = k_d / K_p$
P	T/L	∞	0
PI	$0.9 T/L$	$\frac{L}{0.3}$	0
PID	$1.2 T/L$	$2L$	$0.5L$

Table shows that TF of controller is

$$G_c(s) = k_p \left(1 + \frac{1}{T_i s} + T_d s \right)$$

$$= 1.2 \left(\frac{1}{L} \right) \left(1 + \frac{1}{2L s} + 0.5L s \right)$$

$$G_c(s) = 0.6 T \frac{(s + \frac{1}{L})^2}{s}$$

PID controller has a pole at origin and a double zero at $s = -\frac{1}{L}$

Second method: Dynamic model of the system is known

Type of controller	k_p	T_i	T_d
P	$0.5 K_{cr}$	∞	0
PI	$0.45 K_{cr}$	$\frac{P_{cr}}{1.2}$	0
PID	$0.6 K_{cr}$	$\frac{P_{cr}}{2}$	$\frac{P_{cr}}{8}$

controller has TF

$$G_c(s) = k_p \left(1 + \frac{1}{T_i s} + T_d s \right)$$

$$= 0.6 K_{cr} \left(1 + \frac{1}{0.5 P_{cr} s} + 0.125 P_{cr} s \right)$$

$$G_c(s) = 0.075 K_{cr} P_{cr} \frac{(s + \frac{4}{P_{cr}})^2}{s}$$

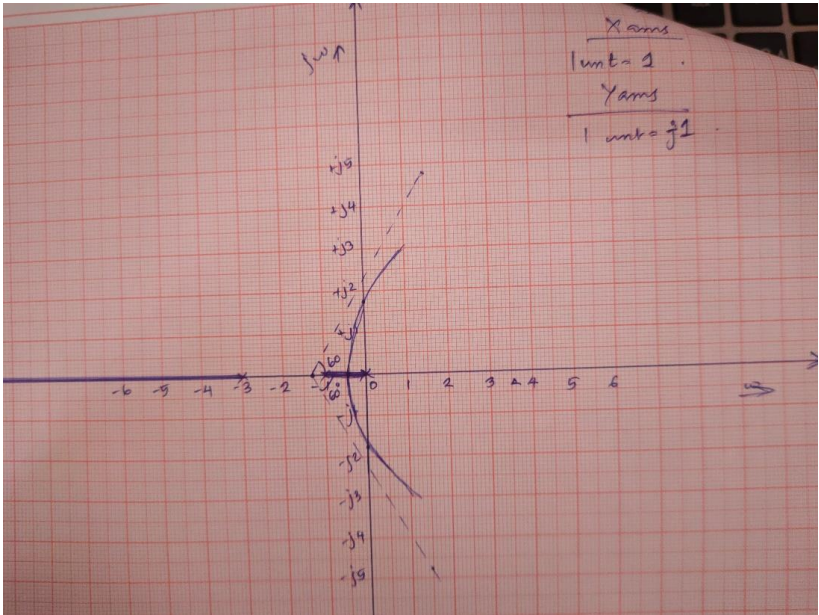
PID controller has pole at origin & a double zero at $s = -\frac{4}{P_{cr}}$

Q16

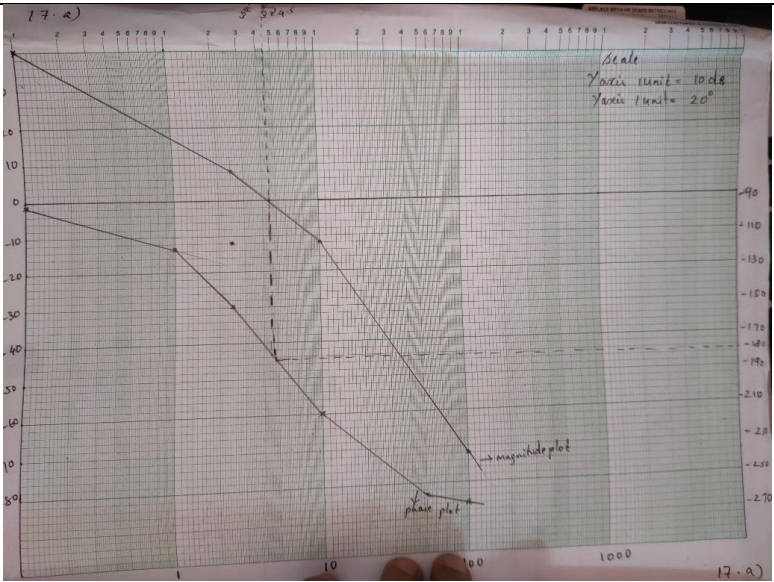
a) A unity control feedback system has an open loop transfer function

$$G(s) = \frac{K}{s(s+1)(s+3)}$$

Draw the root locus for the system and determine the value of K for marginal stability.

		b) Explain the function of PID controllers in any closed loop control system
	Ans	a) A unity control feedback system has an open loop transfer function $G(s) = \frac{K}{s(s+1)(s+3)}$  Breakaway points $s = -0.45$ Angle of asymptotes = 60, 180 and 300 degree Centroid = -1.33 The value of $K=12$ for marginal stability The value of $K =12$ for marginal stability b)

		<u>PID controller</u> o/p $u(t)$ is proportional to $e(t) + \int e(t)dt + \frac{d}{dt} e(t)$ $u(t) \propto \left[e(t) + \int e(t)dt + \frac{d}{dt} e(t) \right]$ $= k_p e(t) + k_i \int e(t)dt + k_d \frac{d}{dt} e(t)$ $U(s) = k_p E(s) + k_i \frac{E(s)}{s} + k_d s E(s)$ $TF = \frac{U(s)}{E(s)} = k_p + \frac{k_i}{s} + k_d s$ <u>PID controller</u> stabilises the gain, reduces steady state error and peak overshoot of the s/m.
Module 4		
Q17	a)	For the given open loop transfer function $G(s) = \frac{10}{s(1 + 0.4s)(1 + 0.1s)}$ Draw the bode plot using asymptotic approach; determine the phase cross over frequency, gain cross over frequency and comment on the stability of the system.
	b)	Define minimum phase and non-minimum phase systems with examples
Ans	a)	For the given open loop transfer function $G(s) = \frac{10}{s(1 + 0.4s)(1 + 0.1s)}$ The phase cross over frequency – 5 rad/sec Gain over frequency – 5 rad/sec Stability of the system – marginally stable

		 b) Minimum phase systems: It is a system in which poles and zeros will not lie on the right side of S plane. Ex: $G(s) = \frac{s+1}{s^2+s+1}$ Non-minimum phase systems: It is a system in which some of the poles and zeros may lie on the right side of s plane. In particular zeros lie on the right side of s plane Ex: $G(s) = \frac{s-1}{s^2+s+1}$
Q18	a)	Sketch the polar plot of a unity feedback control system having an open loop transfer function $G(s) = \frac{1}{s(1+s)(1+2s)}$ Determine the gain margin and phase margin of the above system and comment on the stability of the system b) What is the relationship between phase cross over frequency and gain cross over frequency for a stable system? How does the gain margin and phase margin values vary with these frequencies?
Ans	a)	Plotting polar plot

SOLUTION

Given that, $G(s) = 1/s(1+s)(1+2s)$

Put $s = j\omega$.

$$\therefore G(j\omega) = \frac{1}{j\omega(1+j\omega)(1+j2\omega)}$$

The corner frequencies are $\omega_{c1} = 1/2 = 0.5$ rad/sec and $\omega_{c2} = 1$ rad/sec. The magnitude and phase angle of $G(j\omega)$ are calculated for the corner frequencies and for frequencies around corner frequencies and tabulated in table-1. Using polar to rectangular conversion, the polar coordinates listed in table-1 are converted to rectangular coordinates and tabulated in table-2. The polar plot using polar coordinates is sketched on a polar graph sheet as shown in fig 4.7.1. The polar plot using rectangular coordinates is sketched on an ordinary graph sheet as shown in fig 4.7.2.

$$G(j\omega) = \frac{1}{(j\omega)(1+j\omega)(1+j2\omega)} = \frac{1}{\omega \angle 90^\circ \sqrt{1+\omega^2} \angle \tan^{-1}\omega \sqrt{1+4\omega^2} \angle \tan^{-1}2\omega}$$

$$= \frac{1}{\omega \sqrt{(1+\omega^2)(1+4\omega^2)}} \angle -90^\circ - \tan^{-1}\omega - \tan^{-1}2\omega$$

$$\therefore |G(j\omega)| = \frac{1}{\omega \sqrt{(1+\omega^2)(1+4\omega^2)}} = \frac{1}{\omega \sqrt{1+4\omega^2+\omega^2+4\omega^4}} = \frac{1}{\omega \sqrt{1+5\omega^2+4\omega^4}}$$

$$\angle G(j\omega) = -90^\circ - \tan^{-1}\omega - \tan^{-1}2\omega$$

TABLE-1 : Magnitude and phase of $G(j\omega)$ at various frequencies

ω rad/sec	0.35	0.4	0.45	0.5	0.6	0.7	1.0
$ G(j\omega) $	2.2	1.8	1.5	1.2	0.9	0.7	0.3
$\angle G(j\omega)$ deg	-144	-150	-156	-162	-171	-179.5	-198

TABLE-2 : Real and imaginary part of $G(j\omega)$ at various frequencies

ω rad/sec	0.35	0.4	0.45	0.5	0.6	0.7	1.0
$G_R(j\omega)$	-1.78	-1.56	-1.37	-1.14	-0.89	-0.7	-0.29
$G_I(j\omega)$	-1.29	-0.9	-0.61	-0.37	-0.14	0	0.09

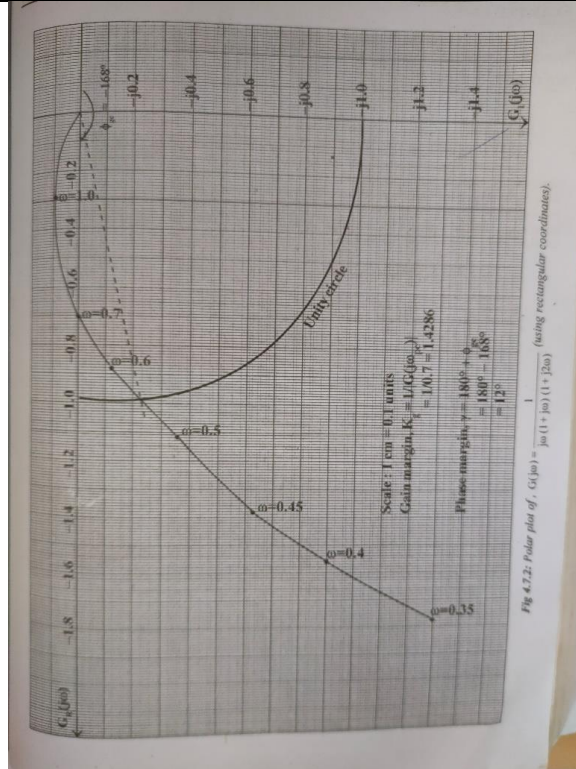
RESULT

Gain margin, $K_g = 1.4286$

Phase margin, $\gamma = +12^\circ$

For a stable system phase cross over frequency should be less than gain cross over frequency.

For a stable system both phase margin and gain margin should be positive.



- b) For a stable system gain cross over frequency should be less than phase cross over frequency for a stable system both phase margin and gain margin should be positive.

Module 5

Q19

- a) Explain the general steps involved in the design of lead compensator using bode plot.
b) Design a phase lead compensator for a unity feedback system given by the open loop transfer function

$$G(s) = \frac{K}{s(s+2)}$$

- i) Phase margin of the system $> 40^\circ$
ii) Velocity error constant is 15/sec

Ans

a)

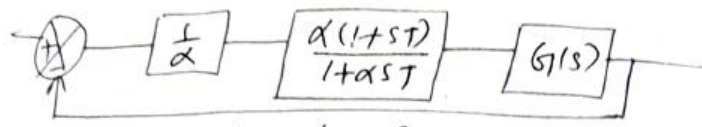
Design of lead comp using Bode plot

- 1) OL gain K is determined to satisfy the requirement of error constant
- 2) Bode plot of uncomp s/m is drawn
- 3) ϕ_m of uncomp s/m is determined from Bode plot.
- 4) Determine the amount of phase angle to be contributed by lead n/w
$$\phi_m = \phi_d + \epsilon - \phi$$

ϕ_m - max phase lead angle of comp
 ϕ_d - desired ϕ_m
 ϕ - ϕ_m of uncomp s/m .
 ϵ - additional phase lead to compensate for shift in ω_{gc}
if $\phi_m > 60$ then 2 cascade comp of each having $\phi_{m\text{new}} = \frac{\phi_m}{2}$
initial value of $\epsilon = 5^\circ$
- 5) Determine TF of comp

6) calculate α using $\alpha = \frac{1 - \sin \phi_m}{1 + \sin \phi_m}$
 from bode plot determine the freq at which mag of $G(j\omega)$ is $-20 \log \frac{1}{\sqrt{\alpha}}$. This freq is ω_m
 calculate T from the relation
 $\omega_m = \frac{1}{T\sqrt{\alpha}}$
 $\therefore$ TF of lead comp
 $\frac{s + \frac{1}{2T}}{s + \frac{1}{\alpha T}} = \frac{\alpha(1 + sT)}{1 + s\alpha T}$

7) determine the OLT of comp sys



$$G_o(s) = \frac{1}{\alpha} \cdot \frac{\alpha(1+sT)}{1+\alpha sT} \cdot G_1(s)$$

8) verify design by drawing Bodeplot & find PM.

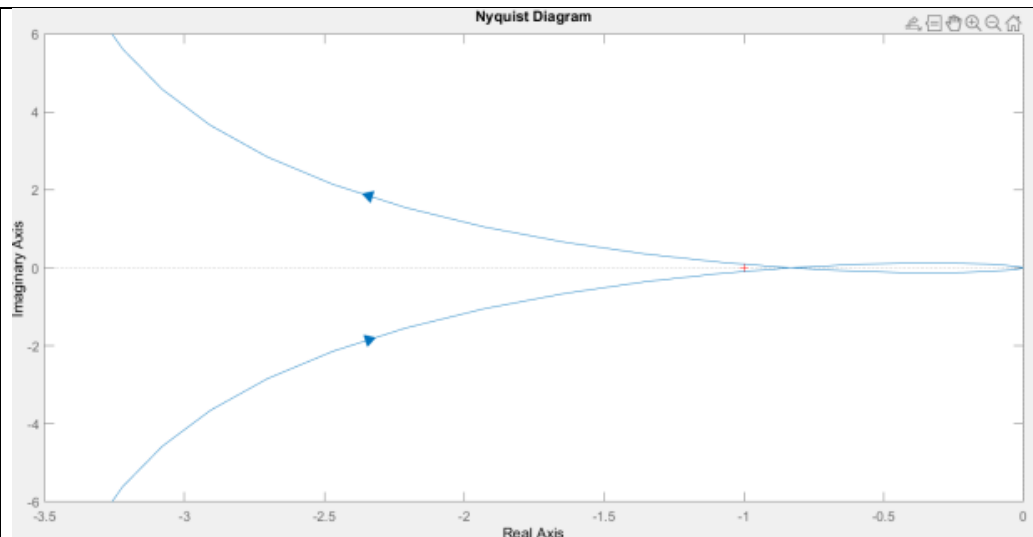
b) A unity feedback system given by the open loop transfer function

$$G(s) = \frac{K}{s(s+2)}$$

GM = infinity

PM = 20.7

Q20	a) The open loop transfer function of a unity feedback system is given by $G(s)H(s) = \frac{5}{s(s+1)(s+2)}$ Draw the Nyquist plot and comment about the stability of the system. b) What is a log magnitude Vs phase plot	
Ans	a) The open loop transfer function of a unity feedback system is given by $G(s)H(s) = 5/[s(s+1)(s+2)]$ At $\omega = 1.42$ rad/sec crosses real Axis at 0. 83 Complete Nyquist plot	

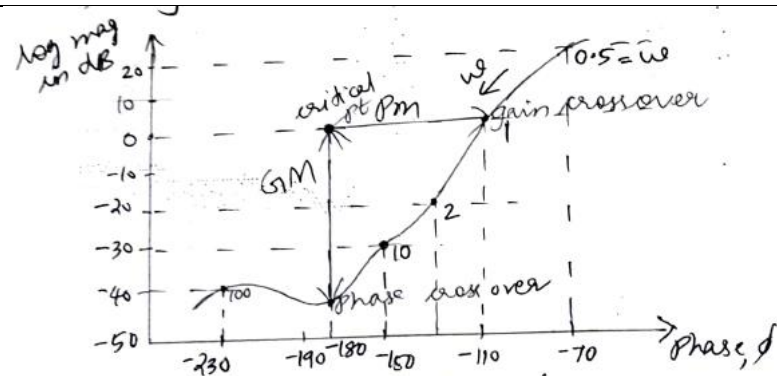


Stability– Closed loop stable

b) Explanation of log Mag Vs phase plot

log magnitude Vs phase plots

log mag Vs phase plot is an alternative approach to freq response representation. In this mag is dB (y axis) Vs phase angle (x axis) is plotted with freq ω as the running parameter. Normally the Bode plot is first obtained & then by reading the values of log mag & phase angle at different freq from Bodeplot, log mag Vs phase plot is constructed. The advantages are relative stability of CL s/m can be determined quickly & compensation can be carried out easily.



- critical pt - intersection of 0dB & -180° axis
- phase cross over is where locus intersects 0dB axis
- gain cross over " -180° axis
- GM - vertical distance in dB measured from phase cross over to critical pt.
- PM - horizontal dist in degrees measured from gain cross over to critical pt.